

- The `py-env` tag specifies third-party packages need to be installed; this acts like *requirements.txt*. Modules such as Bokeh and NumPy have been used in the code given below.
- The standard module can be directly imported in the `py-script` tag; we have imported JSON.
- User defined modules need to be specified under the `py-env` tag, mentioning the path, and further imported in the `py-script` tag. Here we have specified the *utils* file under the `py-env` tag and imported it in the `py-script` tag.

In the `py-script` tag we have saved output in the 'out' id, and later used it in *div*. We can source the Python file similarly.

```
py-script src="./todo.py"> </py-script>
```

Here is the code for `py-env`:

```
<html><head>
<title>Bokeh Example</title>
<meta charset="iso-8859-1">
<link rel="icon" type="image/x-icon" href="./favicon.png">
<script type="text/javascript" src="https://cdn.bokeh.org/bokeh/release/bokeh-2.4.2.min.js"></script>
<script type="text/javascript" src="https://cdn.bokeh.org/bokeh/release/bokeh-gl-2.4.2.min.js"></script>
<script type="text/javascript" src="https://cdn.bokeh.org/bokeh/release/bokeh-widgets-2.4.2.min.js"></script>
<script type="text/javascript" src="https://cdn.bokeh.org/bokeh/release/bokeh-tables-2.4.2.min.js"></script>
<script type="text/javascript" src="https://cdn.bokeh.org/bokeh/release/bokeh-mathjax-2.4.2.min.js"></script>
<script type="text/javascript">
Bokeh.set_log_level("info");
</script>
<link rel="stylesheet" href="https://pyscript.net/alpha/pyscript.css" />
<script defer src="https://pyscript.net/alpha/pyscript.js"></script>
</head>
<body>
<py-env>
- bokeh
- numpy
- paths:
- ./utils.py
</py-env>
<h1>Bokeh Example</h1>
```

```
<py-script output="out">
from utils import now
print("Current time: {}".format(now()))
```

```
</py-script>

<div id="out"></div>
<div id="myplot"></div>
<py-script id="main">
import json
import pyodide
from js import Bokeh, console, JSON
from bokeh.embed import json_item
from bokeh.plotting import figure
from bokeh.resources import CDN
# create a new plot with default tools, using figure
p = figure(plot_width=400, plot_height=400)
# add a circle renderer with x and y coordinates, size,
color, and alpha
p.circle([1, 2, 3, 4, 5], [6, 7, 2, 4, 5], size=15, line_color="navy", fill_color="orange", fill_alpha=0.5)
p_json = json.dumps(json_item(p, "myplot"))
Bokeh.embed.embed_item(JSON.parse(p_json))
</py-script>
</body>
</html>
```

The output for this is shown in Figure 3.

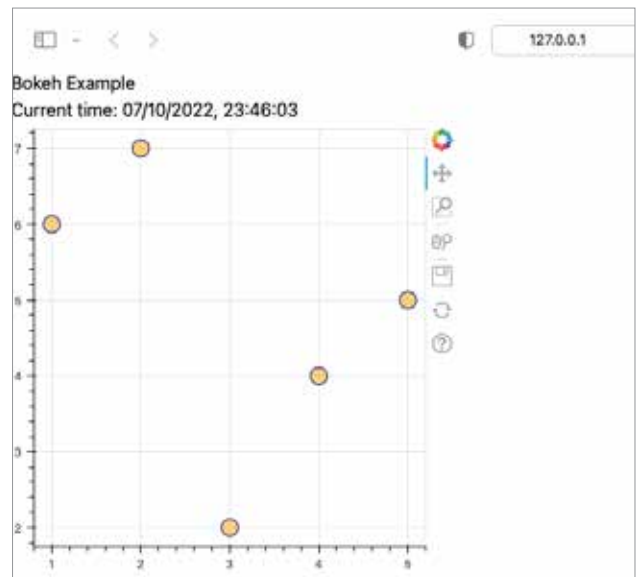


Figure 3: Bokeh example

Python with JavaScript

There is bi-directional communication between Python and JavaScript objects and namespaces. PyScript uses Pyodide's proxy objects to interact with JavaScript objects as WebAssembly doesn't currently allow direct interaction with the document object model (DOM), which would typically give you access to HTML's underlying elements.